大学院ゼミ②

Training data-efficient image transformer &distillation through attention (DeiT)

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目次

論文概要

少ないデータでも高精度に学習できるViT(DeiT)を提案

課題

ViTは高精度だが、大規模データセットと高い計算資源が必要

解決

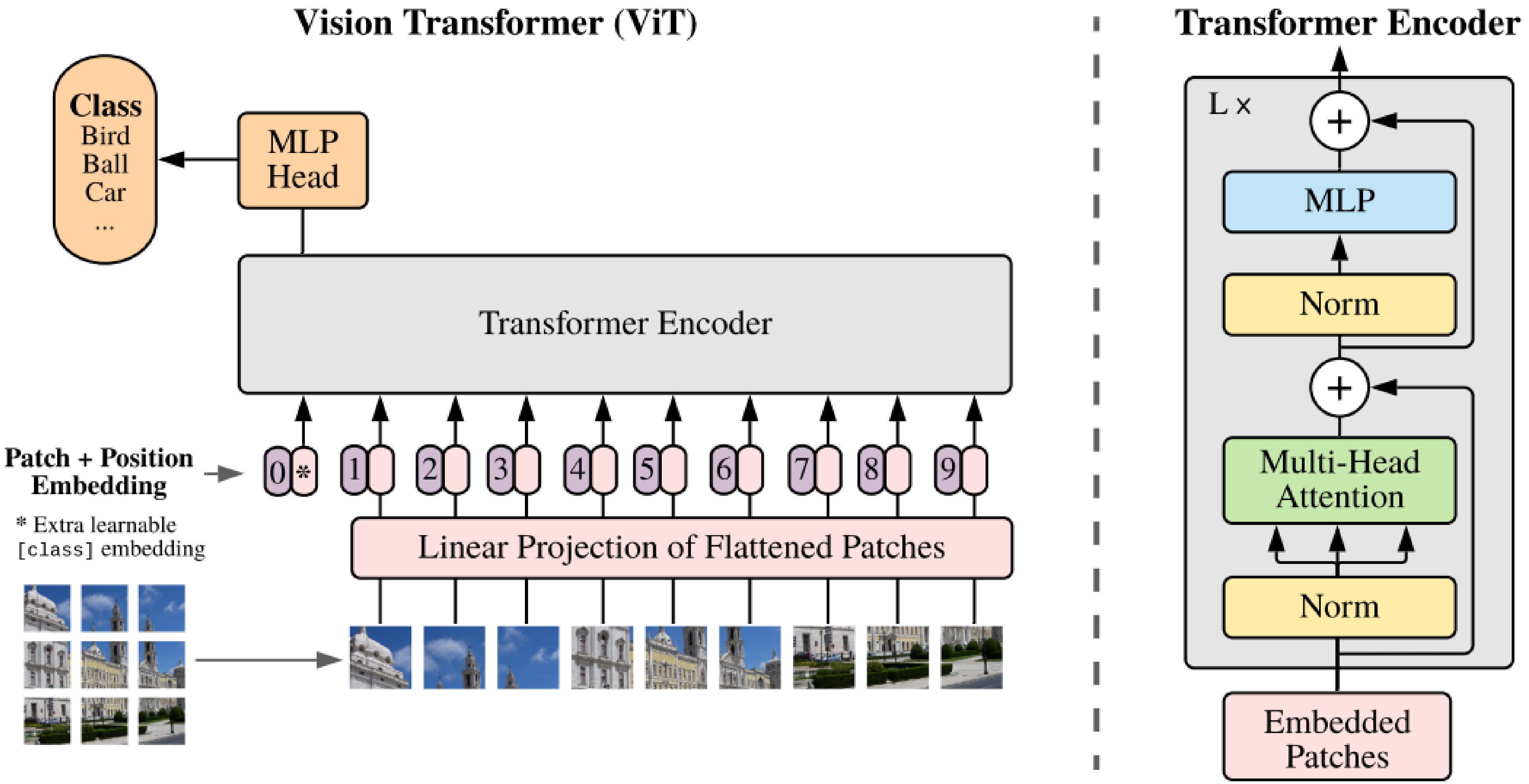
CNNを教師モデルとして利用し、その予測を学習の手がかりにすることで、少ないデータでもViTを高精度に学習する

結果

ImageNetのみで学習し、外部データなしでもCNNに匹敵する精度を達成した

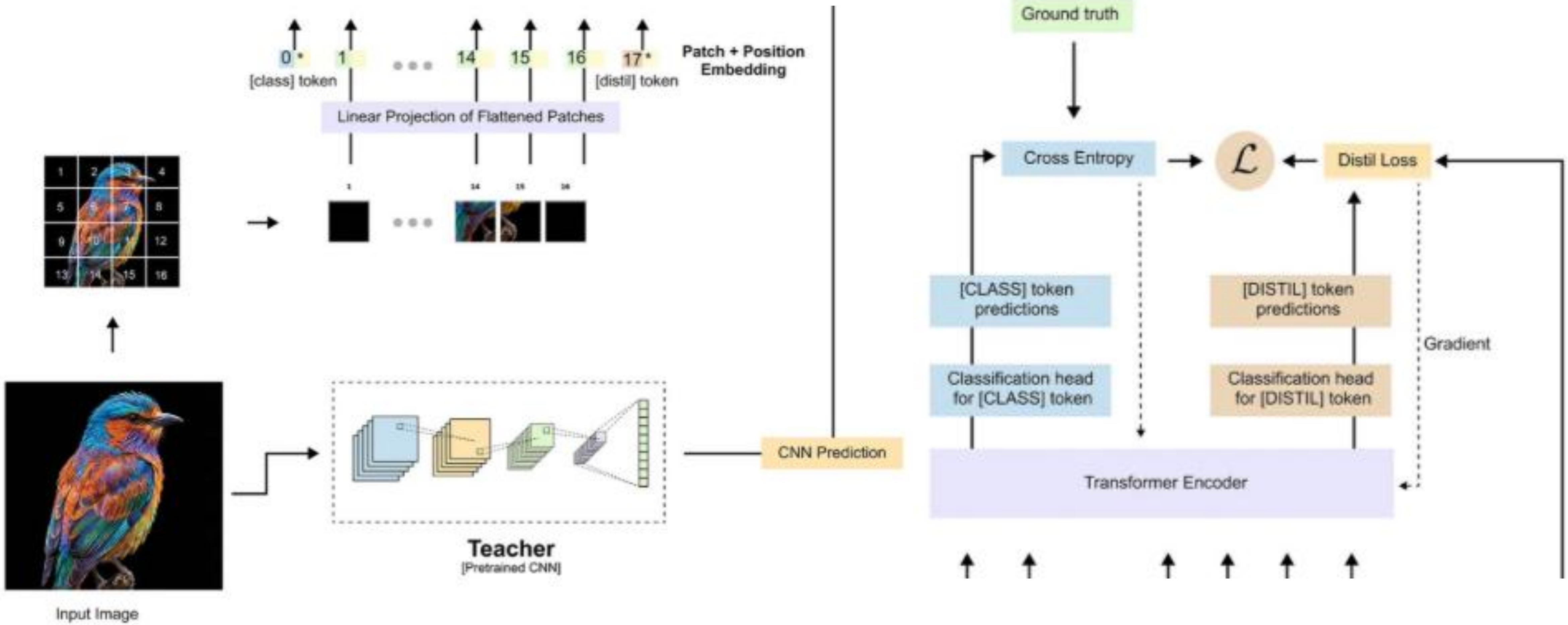
ViTとDeiTの違い

ViT



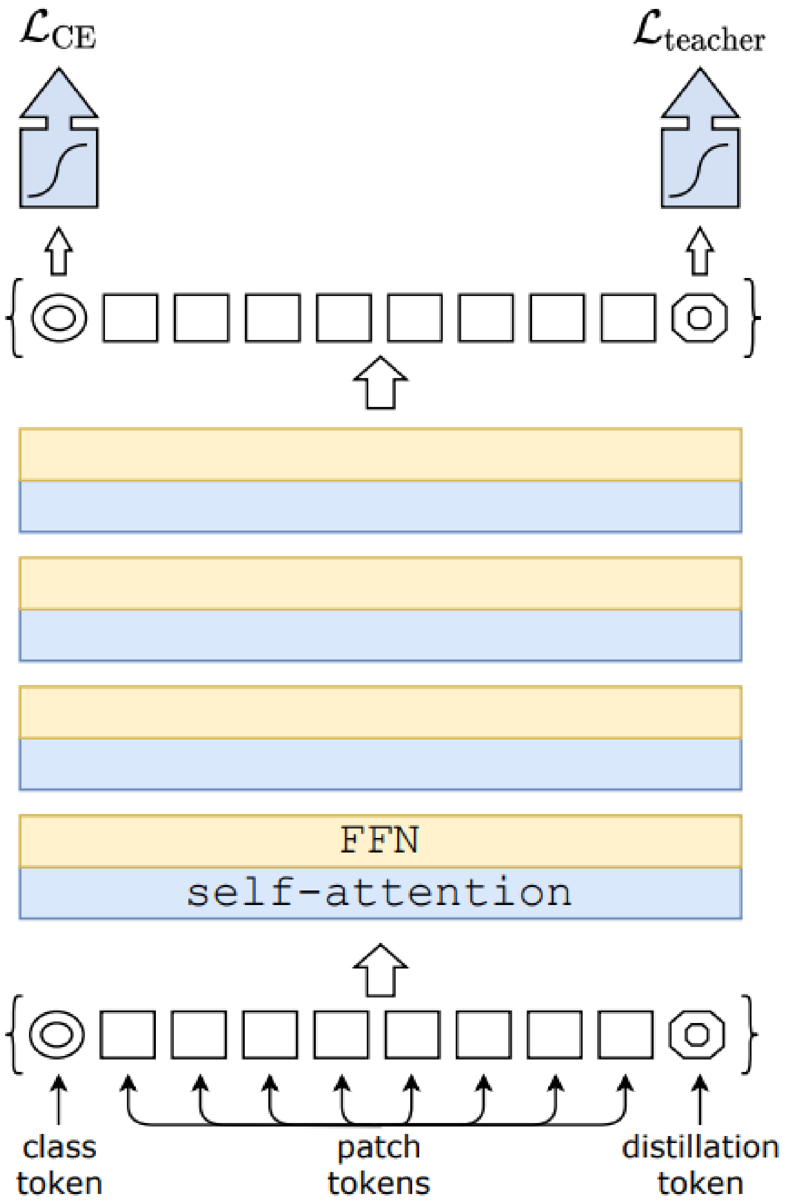
ViTとDeiTの違い

DeiT



https://www.vizuaranewsletter.com/p/knowledge-distillation-and-data-efficient

知識蒸留

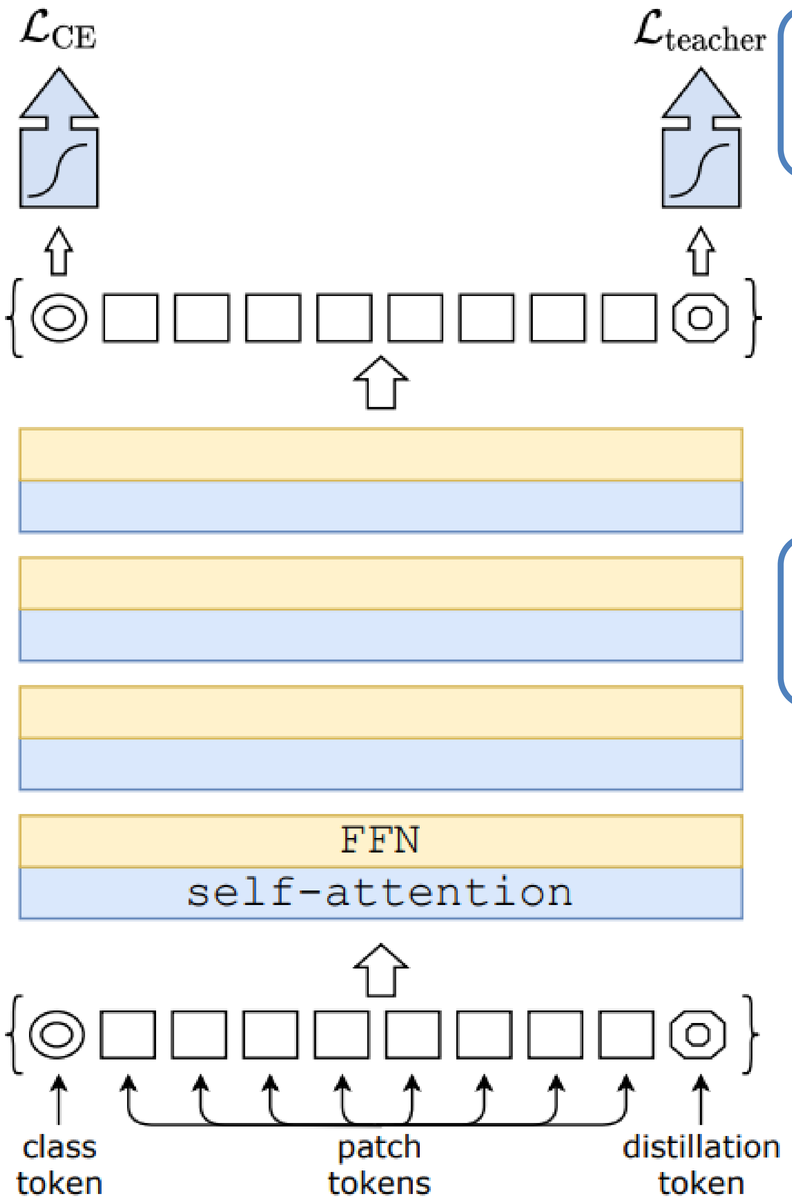


$$L_{CE}(\psi(Z_s), y) \Rightarrow L_{CE} = -\sum_{i=1}^{C} y_i \log p_i^{cls}$$

$$\mathcal{L}_{\text{global}} = (1-\lambda)\mathcal{L}_{\text{CE}}(\psi(Z_{\text{s}}), y) + \lambda \boxed{\tau^2 \text{KL}(\psi(Z_{\text{s}}/\tau), \psi(Z_{\text{t}}/\tau))}.$$

$$L_{teacher} = \boxed{T^2 \sum_{i=1}^{C} p^{(T)}_{teacher,i} \log\left(\frac{p^{(T)}_{teacher,i}}{p^{(T)}_{student,i}}\right)}$$

知識蒸留



Soft distillation

Teacherの確率分布全体を学習するもの

$$\mathcal{L}_{\text{global}} = (1-\lambda)\mathcal{L}_{\text{CE}}(\psi(Z_{\text{s}}), y) + \lambda\tau^2\text{KL}(\psi(Z_{\text{s}}/\tau), \psi(Z_{\text{t}}/\tau)).$$

Hard-label distillation

Teacherが最も高く予測したクラス
だけを学習するもの

$$\mathcal{L}_{\text{global}}^{\text{hardDistill}} = \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_s), y) + \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_s), y_{\text{t}}).$$

実験

①DeiTの蒸留戦略が有効か

②ConvNet(EfficientNet,RegNetY)とViTを精度と推論速度のバランスで比較する

実験

各モデルのパラメータ数

Table 1: Variants of our DeiT architecture. The larger model, DeiT-B, has the same architecture as the ViT-B [15]. The only parameters that vary across models are the embedding dimension and the number of heads, and we keep the dimension per head constant (equal to 64). Smaller models have a lower parameter count, and a faster throughput. The throughput is measured for images at resolution 224×224.

Model	ViT model	embedding dimension	#heads	#layers	#params	training resolution	throughput (im/sec)
DeiT-Ti	N/A	192	3	12	5M	224	2536
DeiT-S	N/A	384	6	12	22M	224	940
DeiT-B	ViT-B	768	12	12	86M	224	292

実験

教師モデルは何が良いか

Table 2: We compare on ImageNet [42] the performance (top-1 acc., %) of the student as a function of the teacher model used for distillation.

Teacher Models	acc.	Student: DeiT-B ⚗ pretrain	↑384
DeiT-B	81.8	81.9	83.1
RegNetY-4GF	80.0	82.7	83.6
RegNetY-8GF	81.7	82.7	83.8
RegNetY-12GF	82.4	83.1	84.1
RegNetY-16GF	82.9	83.1	84.2

実験

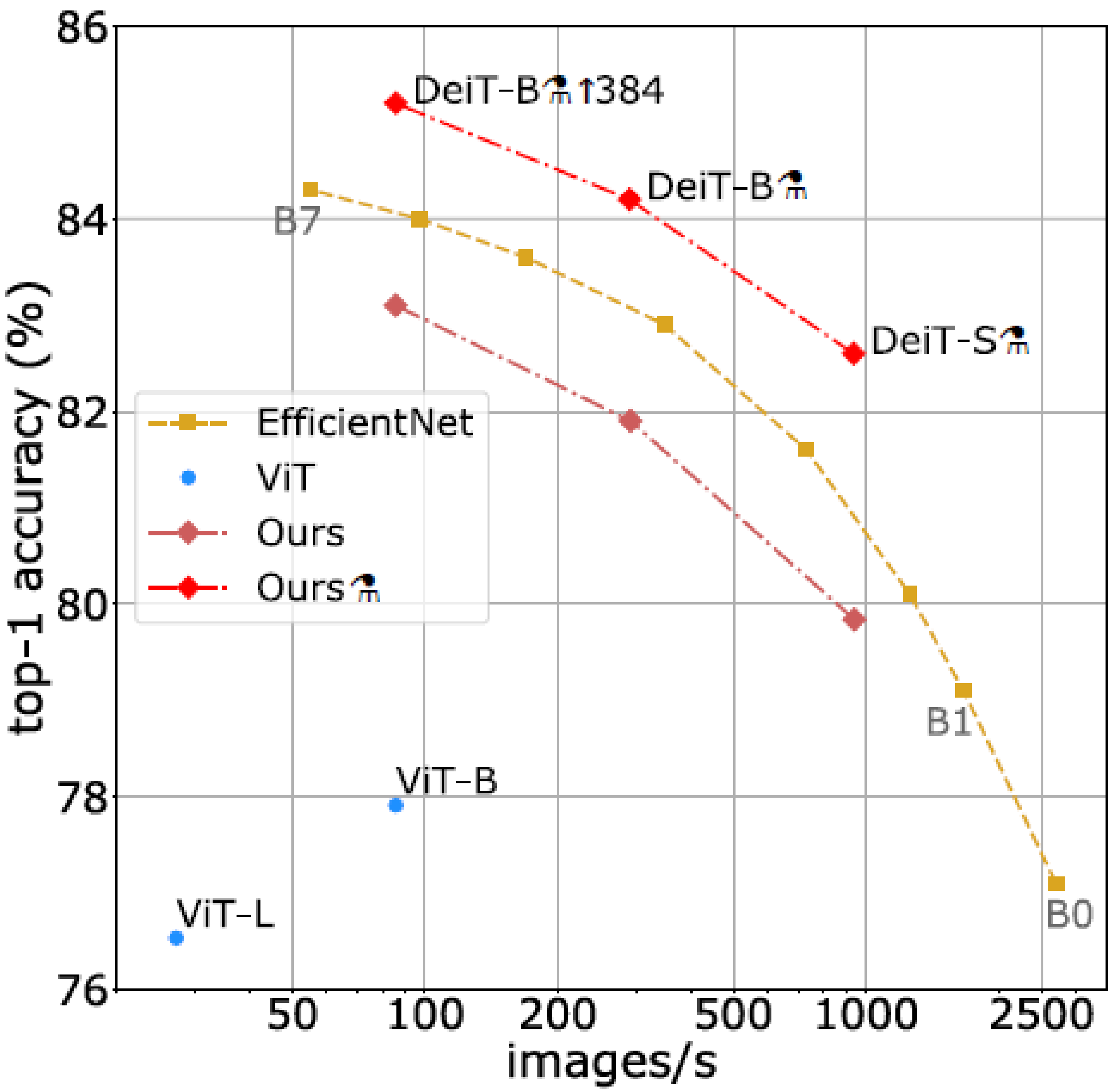
蒸留方法の比較

Table 3: Distillation experiments on Imagenet with DeiT, 300 epochs of pre-training. We report the results for our new distillation method in the last three rows. We separately report the performance when classifying with only one of the class or distillation embeddings, and then with a classifier taking both of them as input. In the last row (class+distillation), the result correspond to the late fusion of the class and distillation classifiers.

	Supervision		ImageNet top-1 (%)			
method ↓	label	teacher	Ti 224	S 224	B 224	B↑384
DeiT– no distillation	✓	✗	72.2	79.8	81.8	83.1
DeiT– usual distillation	✗	soft	72.2	79.8	81.8	83.2
DeiT– hard distillation	✗	hard	74.3	80.9	83.0	84.0
DeiT⚗: class embedding	✓	hard	73.9	80.9	83.0	84.2
DeiT⚗: distil. embedding	✓	hard	74.6	81.1	83.1	84.4
DeiT⚗: class+distillation	✓	hard	74.5	81.2	83.4	84.5

実験

実験結果(評価指標：精度、推論速度)



まとめ

◆外部データなし(事前学習データ不要)で、
高精度に学習するモデルを提案

◆知識蒸留(distillation)により、CNNの
知識をTransformerへ効率的に継承

◆1/234のデータ量で精度差約0.91を達成

補足資料

Methods	ViT-B [15]	DeiT-B
Epochs	300	300
Batch size	4096	1024
Optimizer	AdamW	AdamW
learning rate	0.003	$0.0005 \times \frac{\text{batchsize}}{512}$
Learning rate decay	cosine	cosine
Weight decay	0.3	0.05
Warmup epochs	3.4	5
Label smoothing ε	✗	0.1
Dropout	0.1	✗
Stoch. Depth	✗	0.1
Repeated Aug	✗	✓
Gradient Clip.	✓	✗
Rand Augment	✗	9/0.5
Mixup prob.	✗	0.8
Cutmix prob.	✗	1.0
Erasing prob.	✗	0.25

Table 9: Ingredients and hyper-parameters for our method and Vit-B.

補足資料

Table 6: Datasets used for our different tasks.

Dataset	Train size	Test size	#classes
ImageNet [42]	1,281,167	50,000	1000
iNaturalist 2018 [26]	437,513	24,426	8,142
iNaturalist 2019 [27]	265,240	3,003	1,010
Flowers-102 [38]	2,040	6,149	102
Stanford Cars [30]	8,144	8,041	196
CIFAR-100 [31]	50,000	10,000	100
CIFAR-10 [31]	50,000	10,000	10

Table 7: We compare Transformers based models on different transfer learning task with ImageNet pre-training. We also report results with convolutional architectures for reference.

Model	ImageNet	CIFAR-10	CIFAR-100	Flowers	Cars	iNat-18	iNat-19	im/sec
Grafit ResNet-50 [49]	79.6	–	–	98.2	92.5	69.8	75.9	1226.1
Grafit RegNetY-8GF [49]	–	–	–	99.0	94.0	76.8	80.0	591.6
ResNet-152 [10]	–	–	–	–	–	69.1	–	526.3
EfficientNet-B7 [48]	84.3	98.9	91.7	98.8	94.7	–	–	55.1
ViT-B/32 [15]	73.4	97.8	86.3	85.4	–	–	–	394.5
ViT-B/16 [15]	77.9	98.1	87.1	89.5	–	–	–	85.9
ViT-L/32 [15]	71.2	97.9	87.1	86.4	–	–	–	124.1
ViT-L/16 [15]	76.5	97.9	86.4	89.7	–	–	–	27.3
DeiT-B	81.8	99.1	90.8	98.4	92.1	73.2	77.7	292.3
DeiT-B↑384	83.1	99.1	90.8	98.5	93.3	79.5	81.4	85.9
DeiT-B⚗	83.4	99.1	91.3	98.8	92.9	73.7	78.4	290.9
DeiT-B⚗↑384	84.4	99.2	91.4	98.9	93.9	80.1	83.0	85.9

補足資料

Network	#param.	image size	throughput (image/s)	ImNet top-1	Real top-1	V2 top-1
Convnets						
ResNet-18 [21]	12M	224^2	4458.4	69.8	77.3	57.1
ResNet-50 [21]	25M	224^2	1226.1	76.2	82.5	63.3
ResNet-101 [21]	45M	224^2	753.6	77.4	83.7	65.7
ResNet-152 [21]	60M	224^2	526.4	78.3	84.1	67.0
RegNetY-4GF [40]⋆	21M	224^2	1156.7	80.0	86.4	69.4
RegNetY-8GF [40]⋆	39M	224^2	591.6	81.7	87.4	70.8
RegNetY-16GF [40]⋆	84M	224^2	334.7	82.9	88.1	72.4
EfficientNet-B0 [48]	5M	224^2	2694.3	77.1	83.5	64.3
EfficientNet-B1 [48]	8M	240^2	1662.5	79.1	84.9	66.9
EfficientNet-B2 [48]	9M	260^2	1255.7	80.1	85.9	68.8
EfficientNet-B3 [48]	12M	300^2	732.1	81.6	86.8	70.6
EfficientNet-B4 [48]	19M	380^2	349.4	82.9	88.0	72.3
EfficientNet-B5 [48]	30M	456^2	169.1	83.6	88.3	73.6
EfficientNet-B6 [48]	43M	528^2	96.9	84.0	88.8	73.9
EfficientNet-B7 [48]	66M	600^2	55.1	84.3	-	-
EfficientNet-B5 RA [12]	30M	456^2	96.9	83.7	-	-
EfficientNet-B7 RA [12]	66M	600^2	55.1	84.7	-	-
KDforAA-B8	87M	800^2	25.2	85.8	-	-
Transformers						
ViT-B/16 [15]	86M	384^2	85.9	77.9	83.6	-
ViT-L/16 [15]	307M	384^2	27.3	76.5	82.2	-
DeiT-Ti	5M	224^2	2536.5	72.2	80.1	60.4
DeiT-S	22M	224^2	940.4	79.8	85.7	68.5
DeiT-B	86M	224^2	292.3	81.8	86.7	71.5
DeiT-B↑384	86M	384^2	85.9	83.1	87.7	72.4
DeiT-Ti⚗	6M	224^2	2529.5	74.5	82.1	62.9
DeiT-S⚗	22M	224^2	936.2	81.2	86.8	70.0
DeiT-B⚗	87M	224^2	290.9	83.4	88.3	73.2
DeiT-Ti⚗ / 1000 epochs	6M	224^2	2529.5	76.6	83.9	65.4
DeiT-S⚗ / 1000 epochs	22M	224^2	936.2	82.6	87.8	71.7
DeiT-B⚗ / 1000 epochs	87M	224^2	290.9	84.2	88.7	73.9
DeiT-B⚗ ↑384	87M	384^2	85.8	84.5	89.0	74.8
DeiT-B⚗ ↑384 / 1000 epochs	87M	384^2	85.8	85.2	89.3	75.2

補足資料

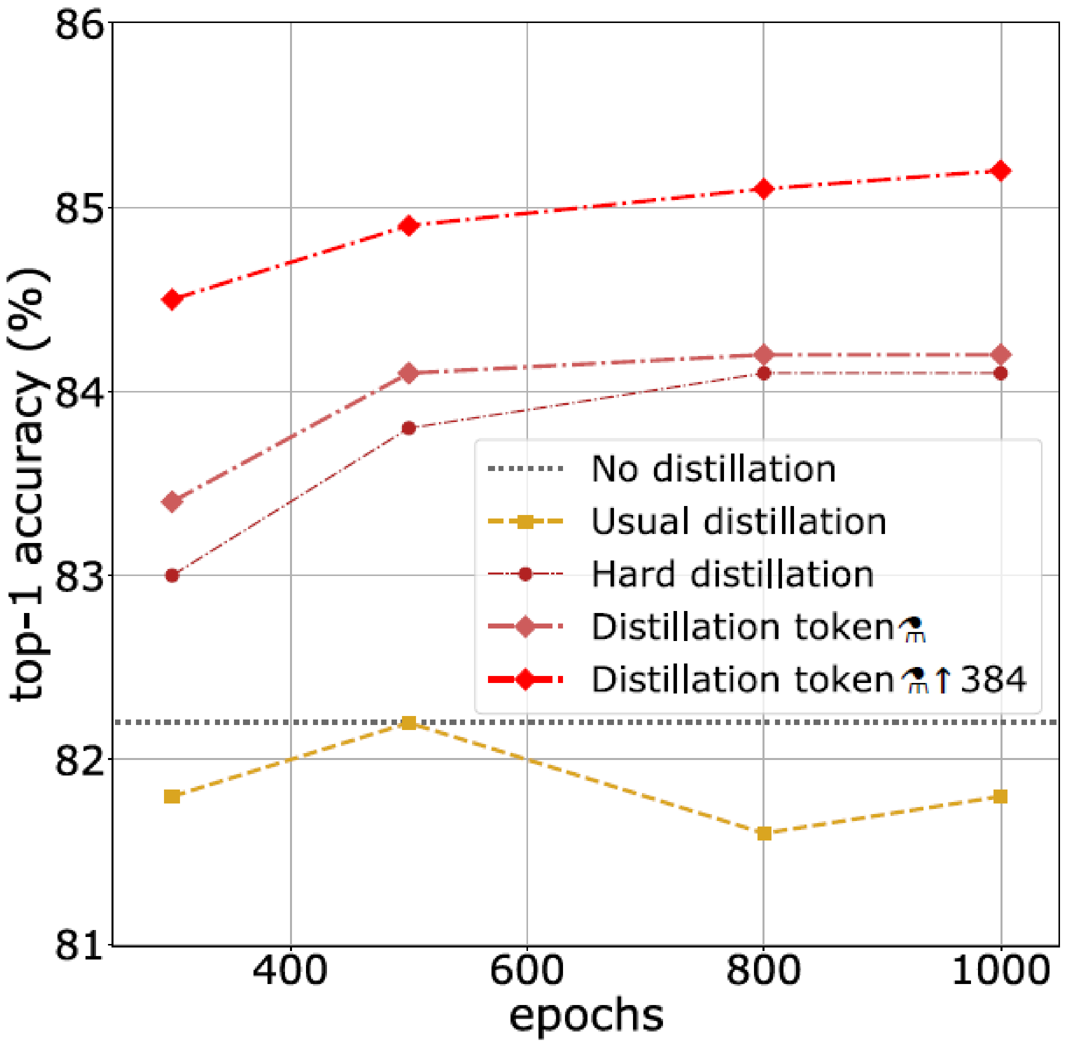


Figure 3: Distillation on ImageNet [42] with DeiT-B: performance as a function of the number of training epochs. We provide the performance without distillation (horizontal dotted line) as it saturates after 400 epochs.